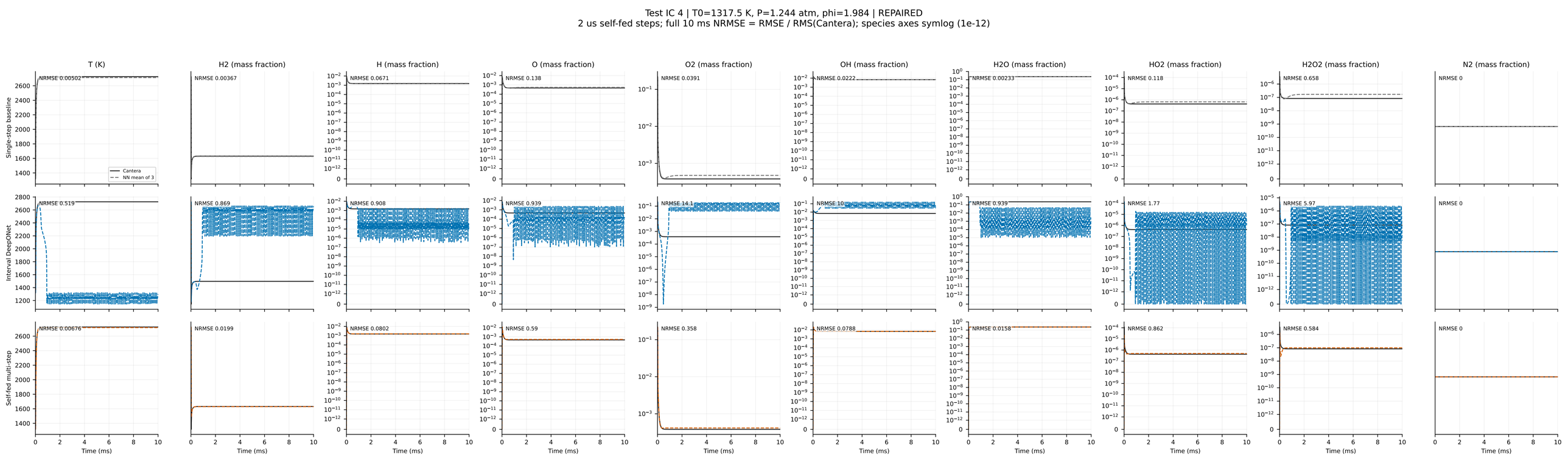
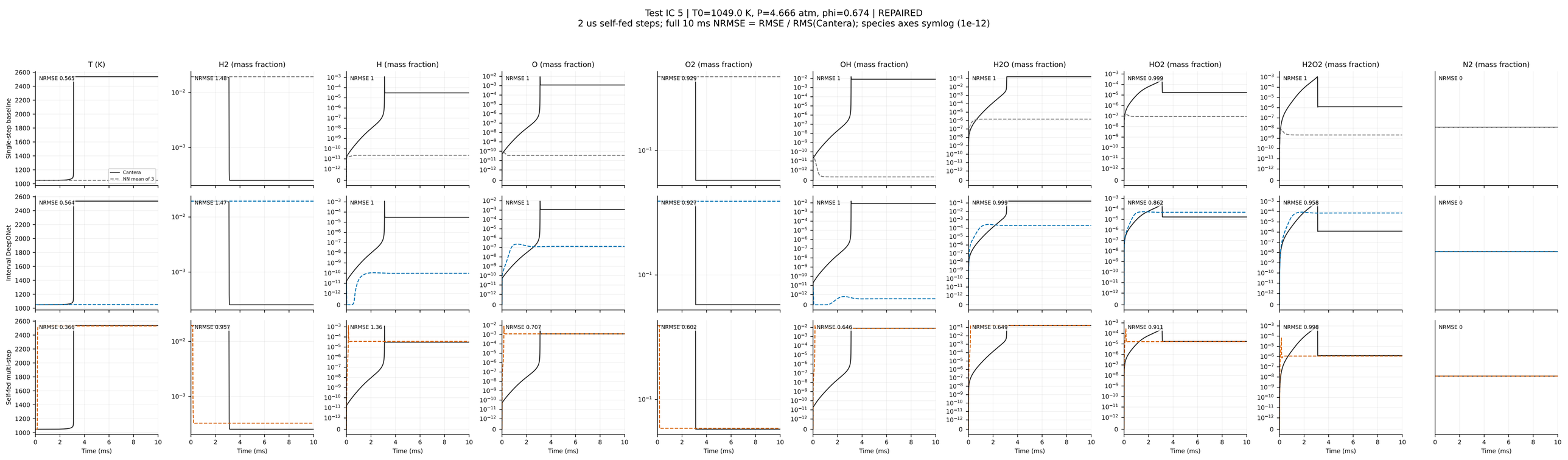
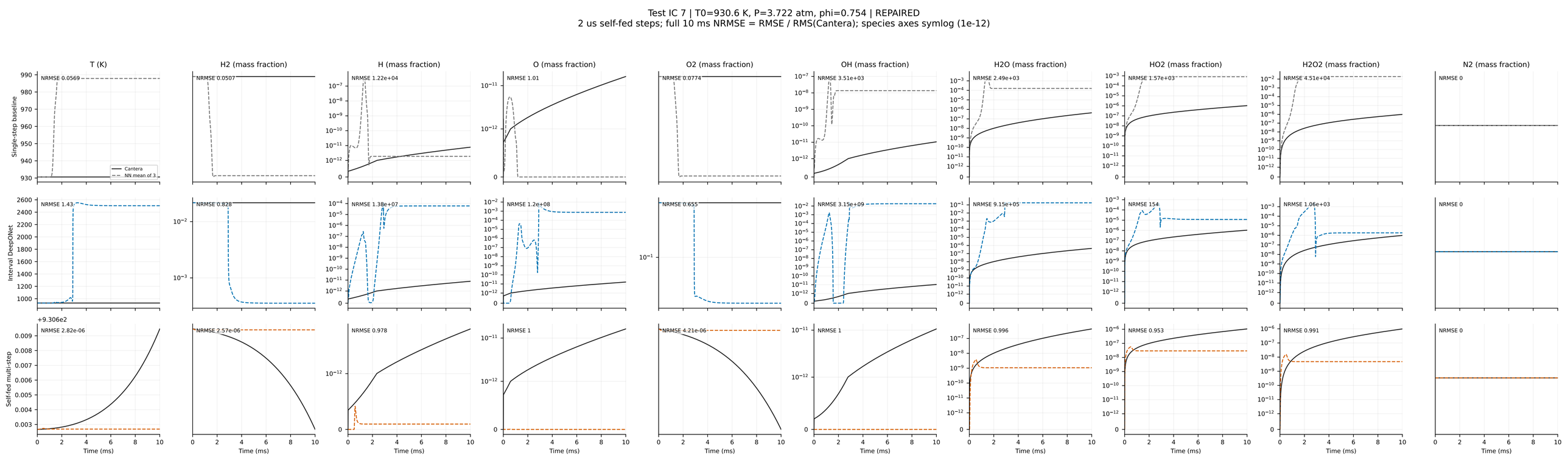
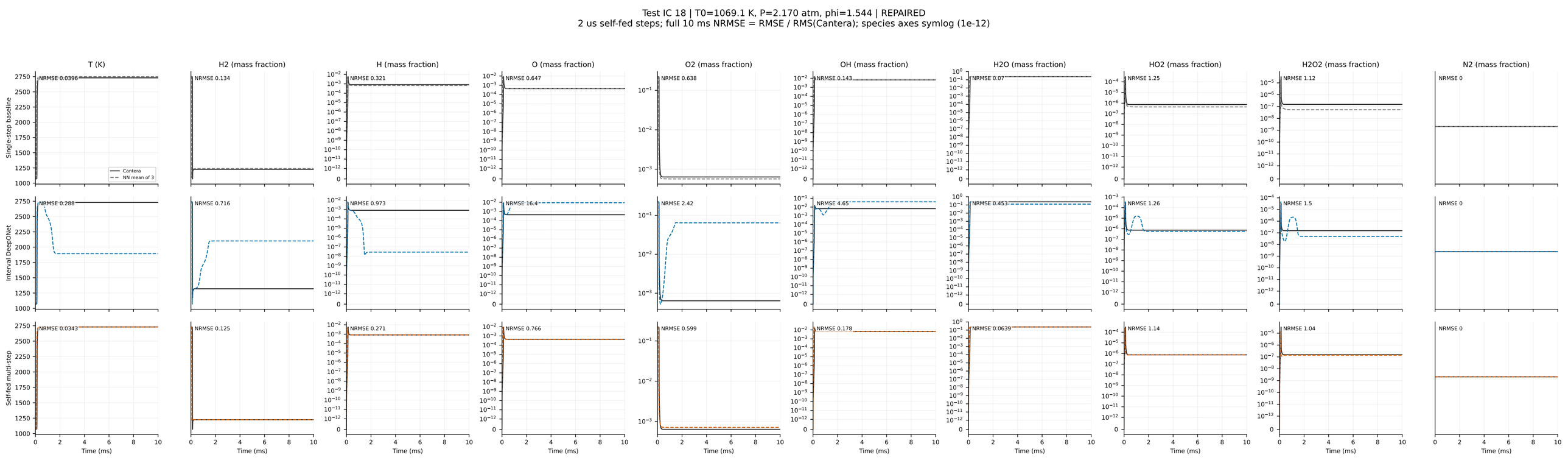
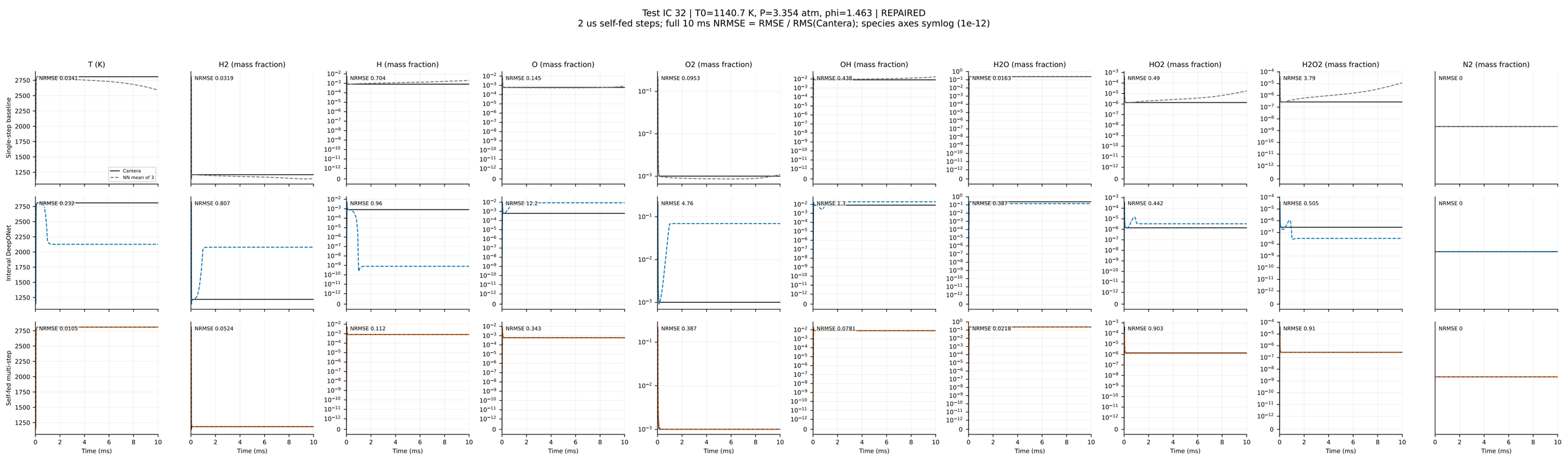
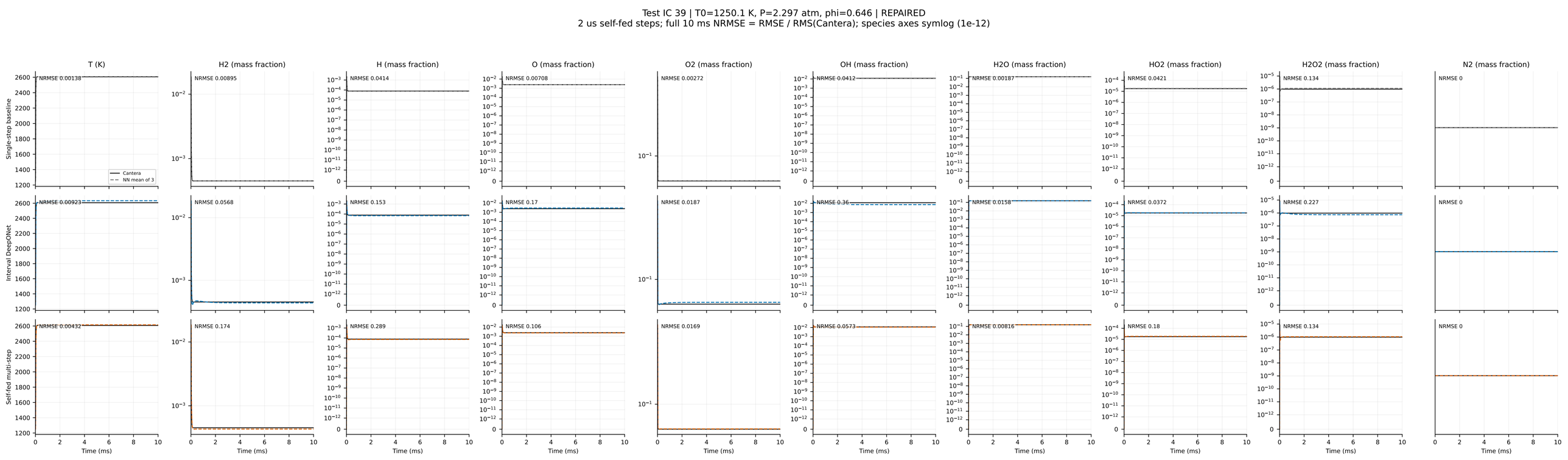

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.00502
NRMSE 0.00367
NRMSE 0.0671
NRMSE 0.138
NRMSE 0.0391
NRMSE 0.0222
NRMSE 0.00233
NRMSE 0.118
NRMSE 0.658
NRMSE 0
NRMSE 0.519
NRMSE 0.869
NRMSE 0.908
NRMSE 0.939
NRMSE 14.1
NRMSE 10
NRMSE 0.939
NRMSE 1.77
NRMSE 5.97
NRMSE 0
NRMSE 0.00676
NRMSE 0.0199
NRMSE 0.0802
NRMSE 0.59
NRMSE 0.358
NRMSE 0.0788
NRMSE 0.0158
NRMSE 0.862
NRMSE 0.584
NRMSE 0
Time (ms)

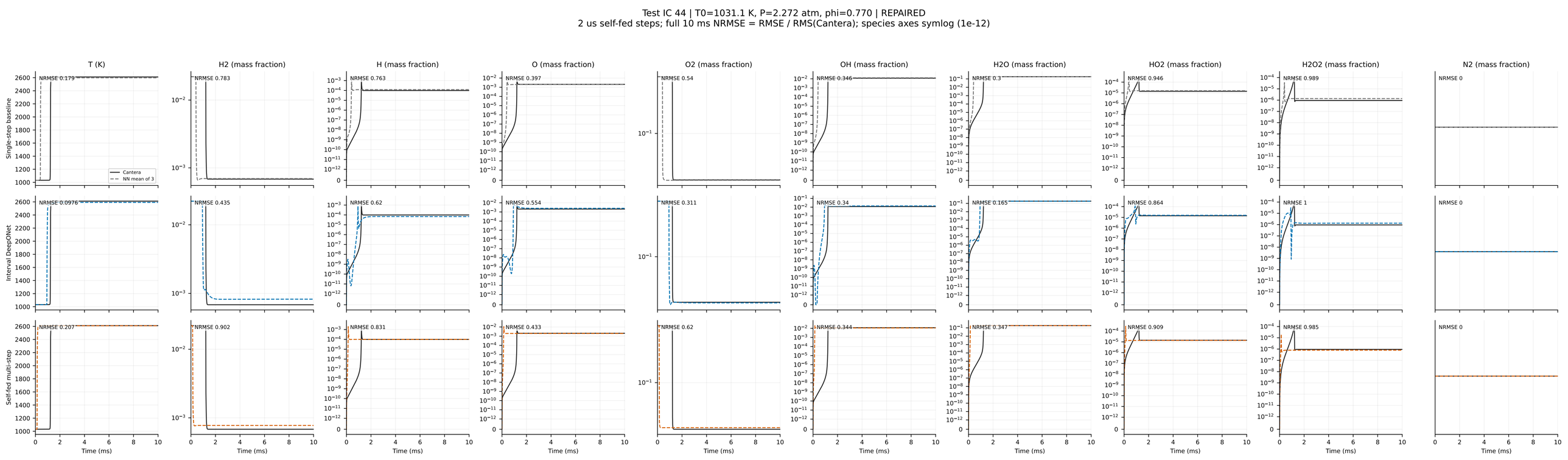

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.565
NRMSE 1.48
NRMSE 1
NRMSE 1
NRMSE 0.929
NRMSE 1
NRMSE 1
NRMSE 0.999
NRMSE 1
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.564
NRMSE 1.47
NRMSE 1
NRMSE 1
NRMSE 0.927
NRMSE 1
NRMSE 0.999
NRMSE 0.862
NRMSE 0.958
NRMSE 0
NRMSE 0.366
NRMSE 0.957
NRMSE 1.36
NRMSE 0.707
NRMSE 0.602
NRMSE 0.646
NRMSE 0.649
NRMSE 0.911
NRMSE 0.998
NRMSE 0
Time (ms)

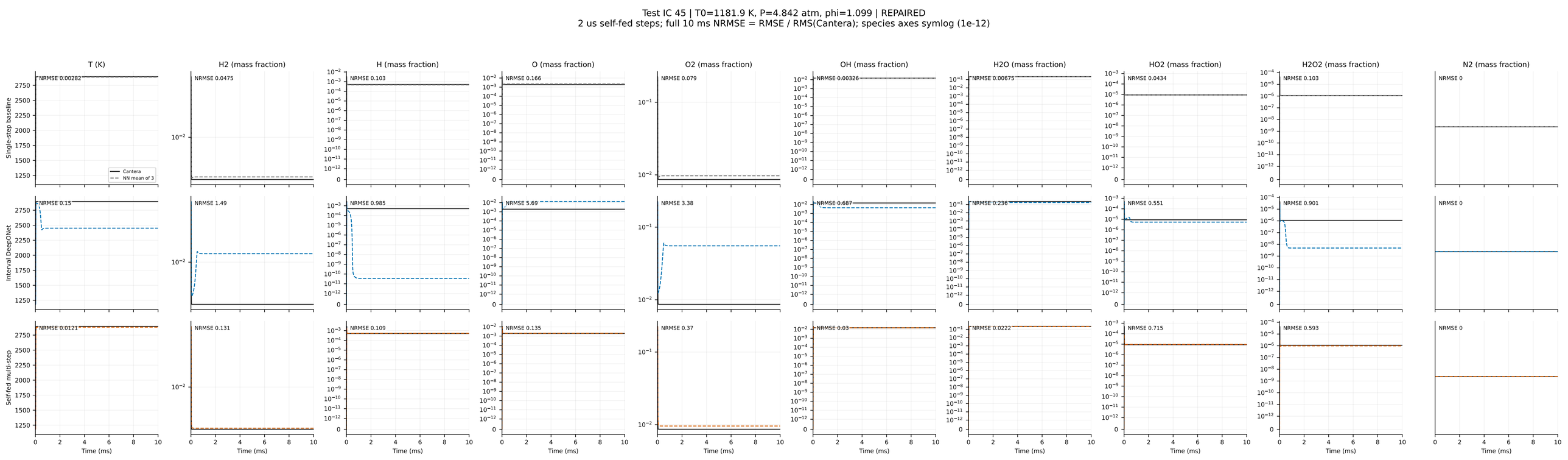

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0569
NRMSE 0.0507
NRMSE 1.22e+04
NRMSE 1.01
NRMSE 0.0774
NRMSE 3.51e+03
NRMSE 2.49e+03
NRMSE 1.57e+03
NRMSE 4.51e+04
NRMSE 0
NRMSE 1.43
NRMSE 0.828
NRMSE 1.38e+07
NRMSE 1.2e+08
NRMSE 0.655
NRMSE 3.15e+09
NRMSE 9.15e+05
NRMSE 154
NRMSE 1.06e+03
NRMSE 0
+9.306e2
NRMSE 2.82e-06
NRMSE 2.57e-06
NRMSE 0.978
NRMSE 1
NRMSE 4.21e-06
NRMSE 1
NRMSE 0.996
NRMSE 0.953
NRMSE 0.991
NRMSE 0
Time (ms)

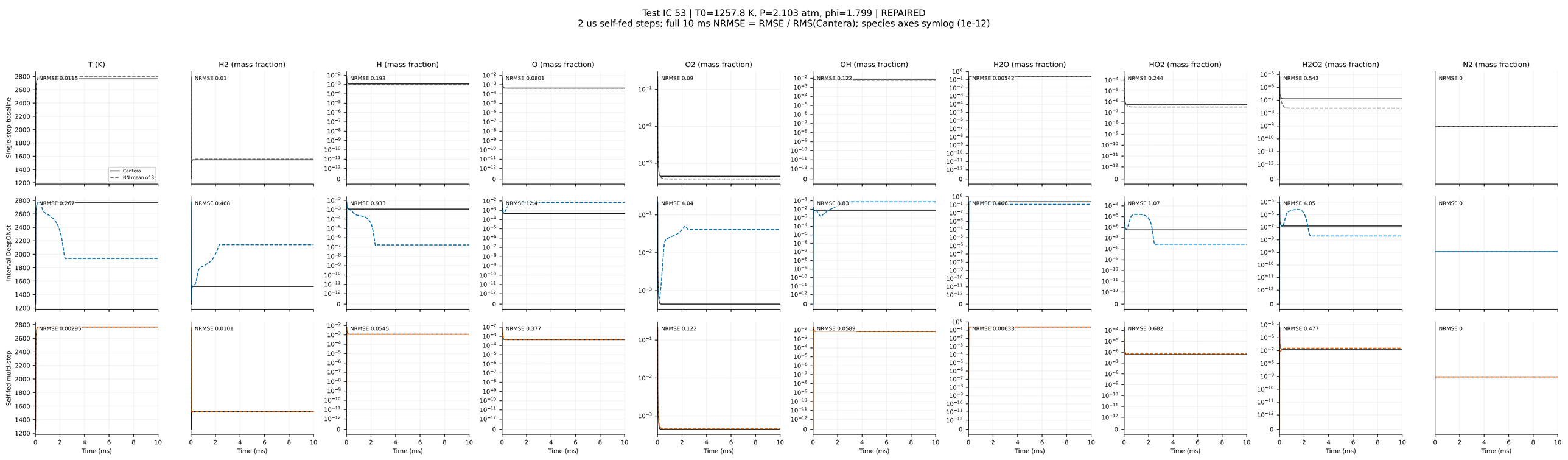

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0396
NRMSE 0.134
NRMSE 0.321
NRMSE 0.647
NRMSE 0.638
NRMSE 0.143
NRMSE 0.07
NRMSE 1.25
NRMSE 1.12
NRMSE 0
NRMSE 0.288
NRMSE 0.716
NRMSE 0.973
NRMSE 16.4
NRMSE 2.42
NRMSE 4.65
NRMSE 0.453
NRMSE 1.26
NRMSE 1.5
NRMSE 0
NRMSE 0.0343
NRMSE 0.125
NRMSE 0.271
NRMSE 0.766
NRMSE 0.599
NRMSE 0.178
NRMSE 0.0639
NRMSE 1.14
NRMSE 1.04
NRMSE 0
Time (ms)

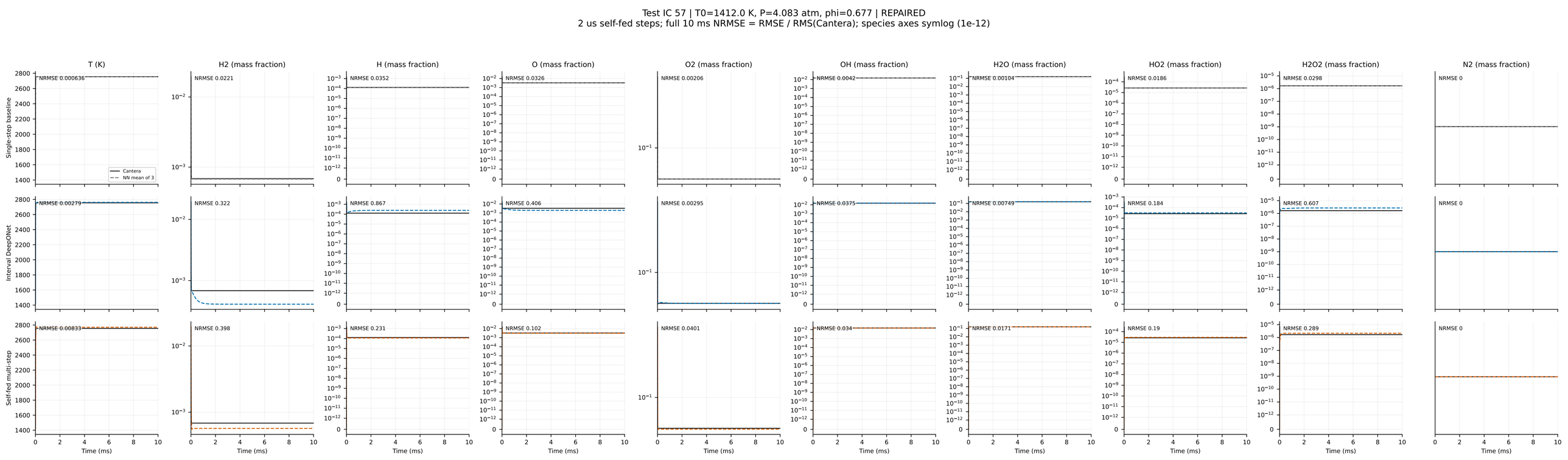

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0341
NRMSE 0.0319
NRMSE 0.704
NRMSE 0.145
NRMSE 0.0953
NRMSE 0.438
NRMSE 0.0163
NRMSE 0.49
NRMSE 3.79
NRMSE 0
NRMSE 0.232
NRMSE 0.807
NRMSE 0.96
NRMSE 12.2
NRMSE 4.76
NRMSE 1.3
NRMSE 0.387
NRMSE 0.442
NRMSE 0.505
NRMSE 0
NRMSE 0.0105
NRMSE 0.0524
NRMSE 0.112
NRMSE 0.343
NRMSE 0.387
NRMSE 0.0781
NRMSE 0.0218
NRMSE 0.903
NRMSE 0.91
NRMSE 0
Cantera
NN mean of 3
Time (ms)

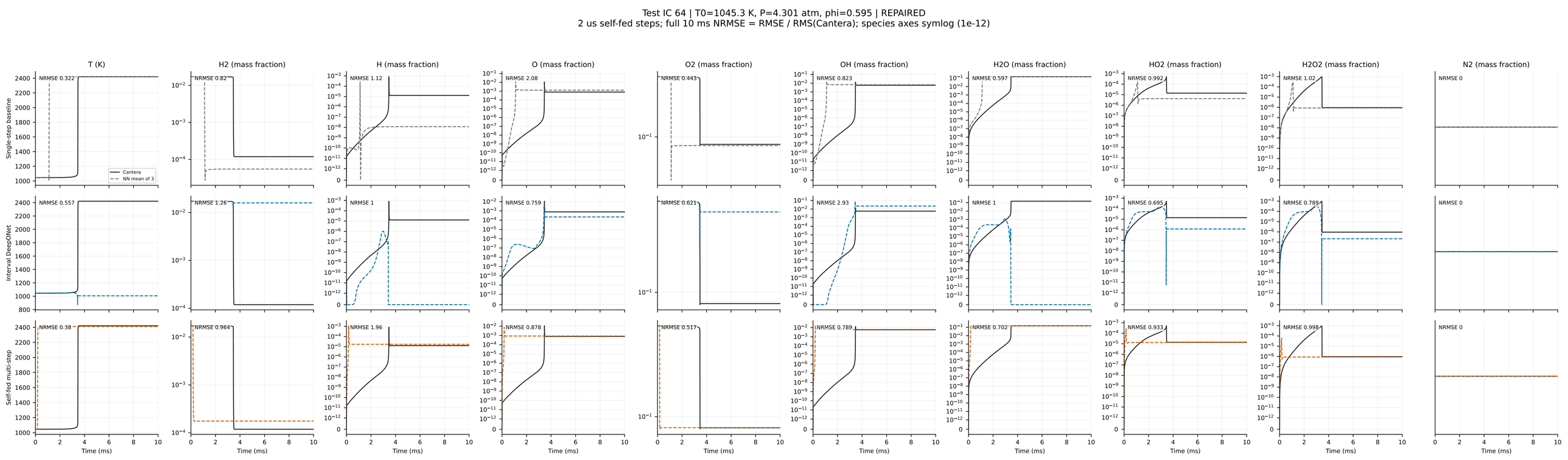

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.00138
NRMSE 0.00895
NRMSE 0.0414
NRMSE 0.00708
NRMSE 0.00272
NRMSE 0.0412
NRMSE 0.00187
NRMSE 0.0421
NRMSE 0.134
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.00923
NRMSE 0.0568
NRMSE 0.153
NRMSE 0.17
NRMSE 0.0187
NRMSE 0.36
NRMSE 0.0158
NRMSE 0.0372
NRMSE 0.227
NRMSE 0
NRMSE 0.00432
NRMSE 0.174
NRMSE 0.289
NRMSE 0.106
NRMSE 0.0169
NRMSE 0.0573
NRMSE 0.00816
NRMSE 0.18
NRMSE 0.134
NRMSE 0
Time (ms)

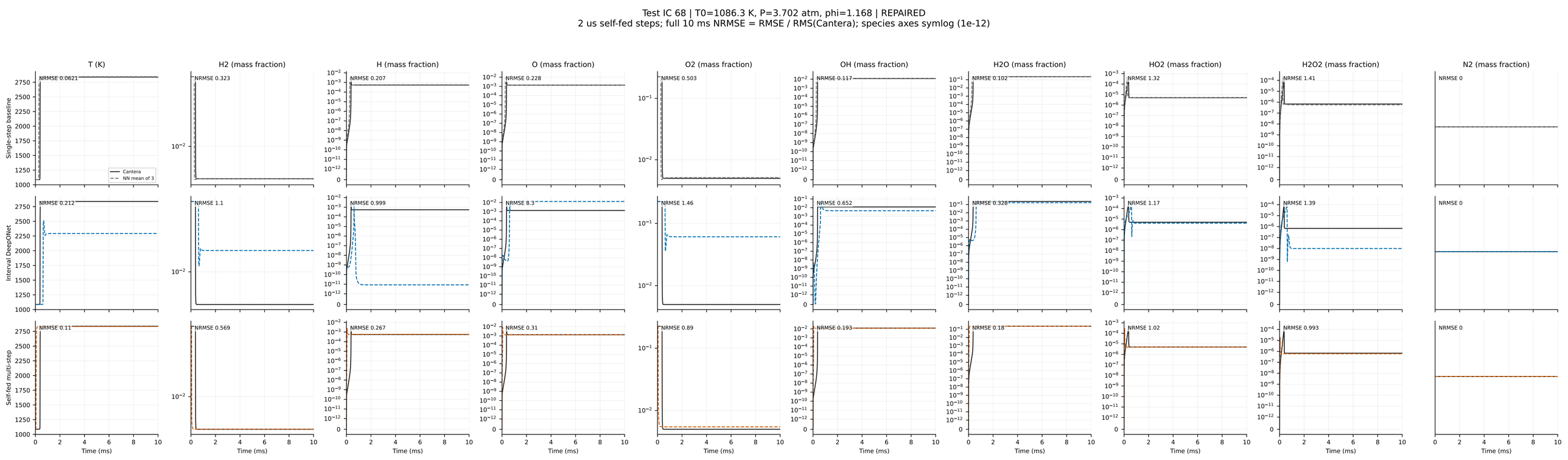

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.179
NRMSE 0.783
NRMSE 0.763
NRMSE 0.397
NRMSE 0.54
NRMSE 0.346
NRMSE 0.3
NRMSE 0.946
NRMSE 0.989
NRMSE 0
NRMSE 0.0976
NRMSE 0.435
NRMSE 0.62
NRMSE 0.554
NRMSE 0.311
NRMSE 0.34
NRMSE 0.165
NRMSE 0.864
NRMSE 1
NRMSE 0
NRMSE 0.207
NRMSE 0.902
NRMSE 0.831
NRMSE 0.433
NRMSE 0.62
NRMSE 0.344
NRMSE 0.347
NRMSE 0.909
NRMSE 0.985
NRMSE 0
Time (ms)

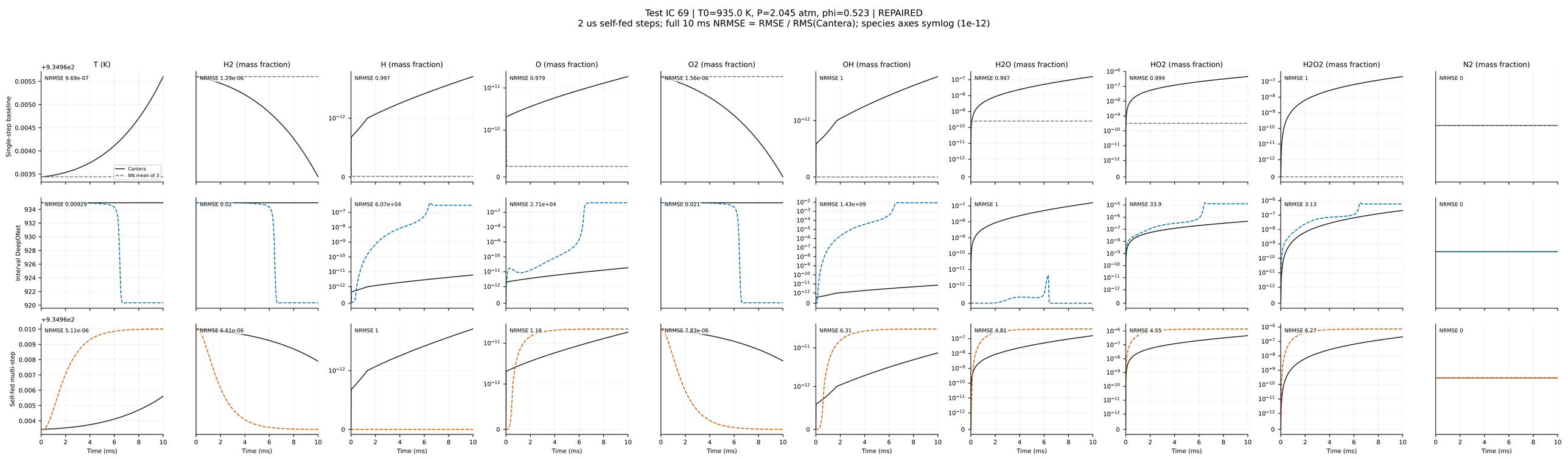

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.00282
NRMSE 0.0475
NRMSE 0.103
NRMSE 0.166
NRMSE 0.079
NRMSE 0.00326
NRMSE 0.00675
NRMSE 0.0434
NRMSE 0.103
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.15
NRMSE 1.49
NRMSE 0.985
NRMSE 5.69
NRMSE 3.38
NRMSE 0.687
NRMSE 0.236
NRMSE 0.551
NRMSE 0.901
NRMSE 0
NRMSE 0.0121
NRMSE 0.131
NRMSE 0.109
NRMSE 0.135
NRMSE 0.37
NRMSE 0.03
NRMSE 0.0222
NRMSE 0.715
NRMSE 0.593
NRMSE 0
Time (ms)

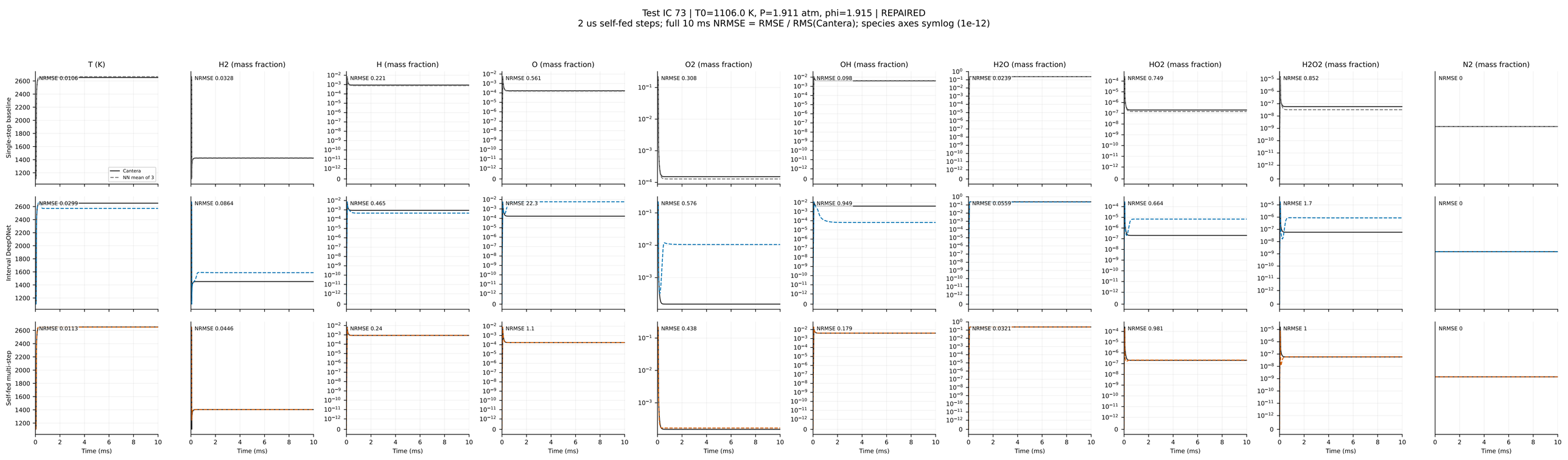

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0115
NRMSE 0.01
NRMSE 0.192
NRMSE 0.0801
NRMSE 0.09
NRMSE 0.122
NRMSE 0.00542
NRMSE 0.244
NRMSE 0.543
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.267
NRMSE 0.468
NRMSE 0.933
NRMSE 12.4
NRMSE 4.04
NRMSE 8.83
NRMSE 0.466
NRMSE 1.07
NRMSE 4.05
NRMSE 0
NRMSE 0.00295
NRMSE 0.0101
NRMSE 0.0545
NRMSE 0.377
NRMSE 0.122
NRMSE 0.0589
NRMSE 0.00633
NRMSE 0.682
NRMSE 0.477
NRMSE 0
Time (ms)

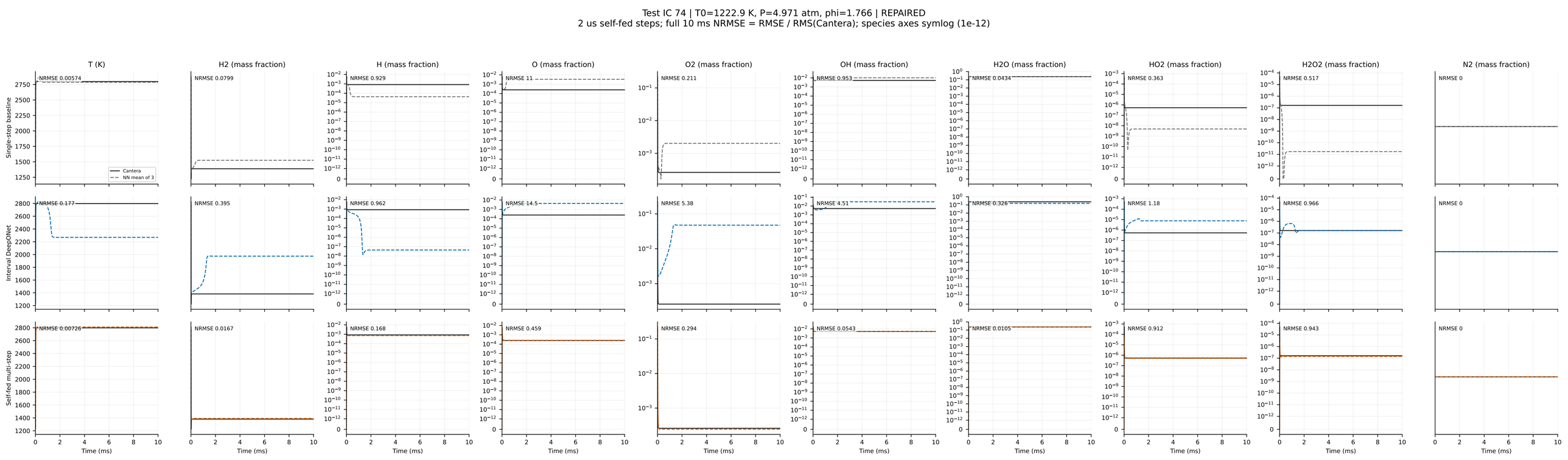

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.000636
NRMSE 0.0221
NRMSE 0.0352
NRMSE 0.0326
NRMSE 0.00206
NRMSE 0.0042
NRMSE 0.00104
NRMSE 0.0186
NRMSE 0.0298
NRMSE 0
NRMSE 0.00279
NRMSE 0.322
NRMSE 0.867
NRMSE 0.406
NRMSE 0.00295
NRMSE 0.0375
NRMSE 0.00749
NRMSE 0.184
NRMSE 0.607
NRMSE 0
NRMSE 0.00833
NRMSE 0.398
NRMSE 0.231
NRMSE 0.102
NRMSE 0.0401
NRMSE 0.034
NRMSE 0.0171
NRMSE 0.19
NRMSE 0.289
NRMSE 0
Cantera
NN mean of 3
Time (ms)

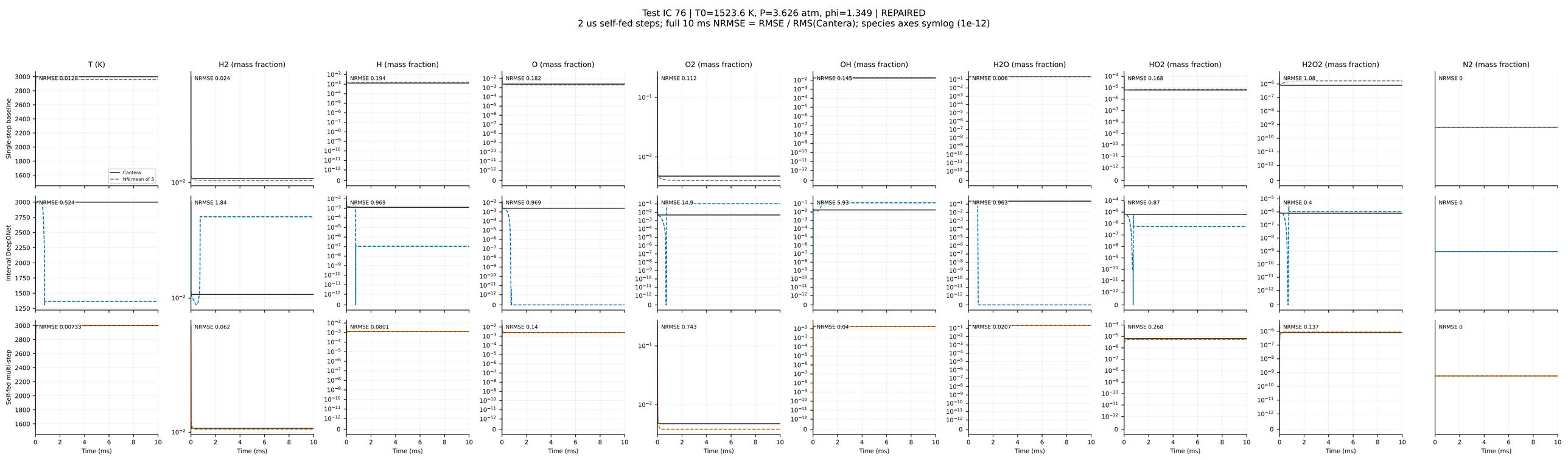

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.322
NRMSE 0.82
NRMSE 1.12
NRMSE 2.08
NRMSE 0.443
NRMSE 0.823
NRMSE 0.597
NRMSE 0.992
NRMSE 1.02
NRMSE 0
NRMSE 0.557
NRMSE 1.26
NRMSE 1
NRMSE 0.759
NRMSE 0.621
NRMSE 2.93
NRMSE 1
NRMSE 0.695
NRMSE 0.789
NRMSE 0
NRMSE 0.38
NRMSE 0.964
NRMSE 1.96
NRMSE 0.878
NRMSE 0.517
NRMSE 0.789
NRMSE 0.702
NRMSE 0.933
NRMSE 0.998
NRMSE 0
Time (ms)

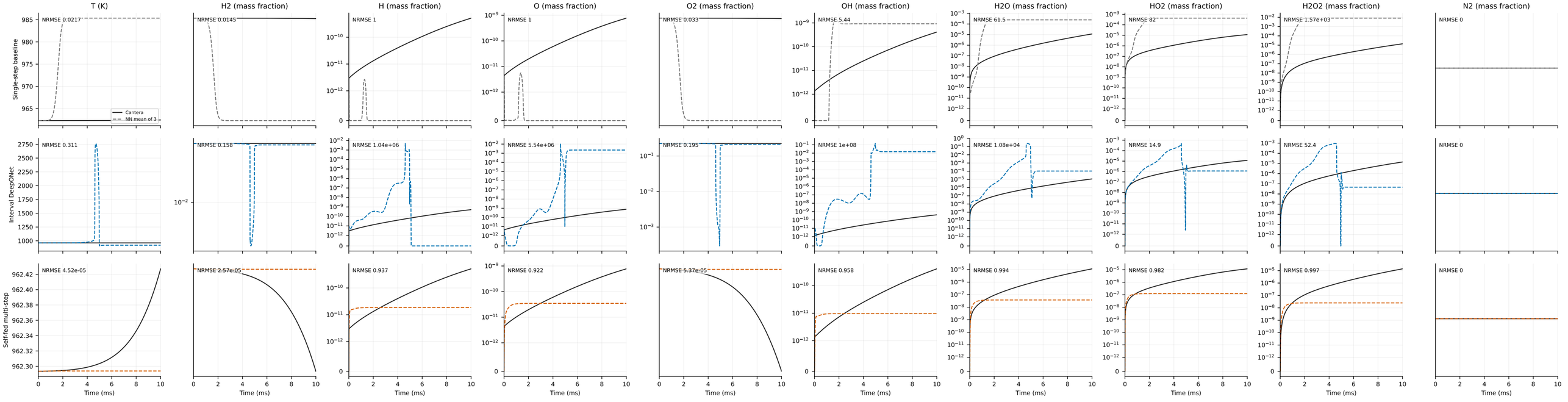

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0621
NRMSE 0.323
NRMSE 0.207
NRMSE 0.228
NRMSE 0.503
NRMSE 0.117
NRMSE 0.102
NRMSE 1.32
NRMSE 1.41
NRMSE 0
NRMSE 0.212
NRMSE 1.1
NRMSE 0.999
NRMSE 8.3
NRMSE 1.46
NRMSE 0.652
NRMSE 0.328
NRMSE 1.17
NRMSE 1.39
NRMSE 0
NRMSE 0.11
NRMSE 0.569
NRMSE 0.267
NRMSE 0.31
NRMSE 0.89
NRMSE 0.193
NRMSE 0.18
NRMSE 1.02
NRMSE 0.993
NRMSE 0
Time (ms)

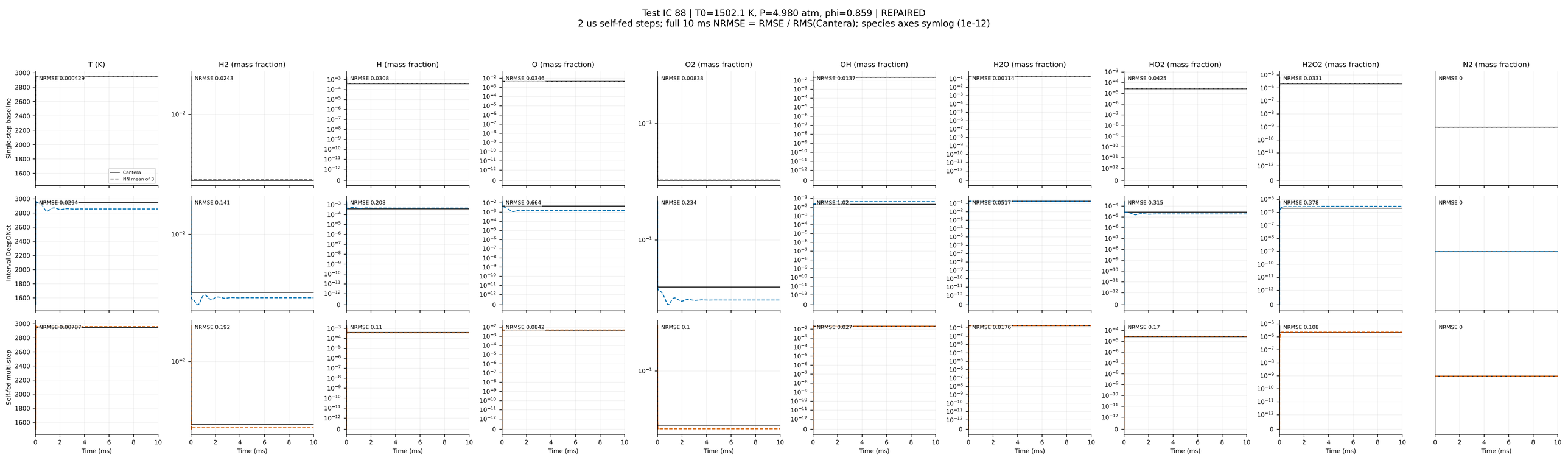

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
+9.3496e2
NRMSE 9.69e-07
NRMSE 1.29e-06
NRMSE 0.997
NRMSE 0.979
NRMSE 1.56e-06
NRMSE 1
NRMSE 0.997
NRMSE 0.999
NRMSE 1
NRMSE 0
Cantera
NN mean of 3
NRMSE 0.00929
NRMSE 0.02
NRMSE 6.07e+04
NRMSE 2.71e+04
NRMSE 0.021
NRMSE 1.43e+09
NRMSE 1
NRMSE 33.9
NRMSE 3.13
NRMSE 0
+9.3496e2
NRMSE 5.11e-06
NRMSE 6.61e-06
NRMSE 1
NRMSE 1.16
NRMSE 7.83e-06
NRMSE 6.31
NRMSE 4.81
NRMSE 4.55
NRMSE 6.27
NRMSE 0
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0106
NRMSE 0.0328
NRMSE 0.221
NRMSE 0.561
NRMSE 0.308
NRMSE 0.098
NRMSE 0.0239
NRMSE 0.749
NRMSE 0.852
NRMSE 0
NRMSE 0.0299
NRMSE 0.0864
NRMSE 0.465
NRMSE 22.3
NRMSE 0.576
NRMSE 0.949
NRMSE 0.0559
NRMSE 0.664
NRMSE 1.7
NRMSE 0
NRMSE 0.0113
NRMSE 0.0446
NRMSE 0.24
NRMSE 1.1
NRMSE 0.438
NRMSE 0.179
NRMSE 0.0321
NRMSE 0.981
NRMSE 1
NRMSE 0
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.00574
NRMSE 0.0799
NRMSE 0.929
NRMSE 11
NRMSE 0.211
NRMSE 0.953
NRMSE 0.0434
NRMSE 0.363
NRMSE 0.517
NRMSE 0
NRMSE 0.177
NRMSE 0.395
NRMSE 0.962
NRMSE 14.5
NRMSE 5.38
NRMSE 4.51
NRMSE 0.326
NRMSE 1.18
NRMSE 0.966
NRMSE 0
NRMSE 0.00726
NRMSE 0.0167
NRMSE 0.168
NRMSE 0.459
NRMSE 0.294
NRMSE 0.0543
NRMSE 0.0105
NRMSE 0.912
NRMSE 0.943
NRMSE 0
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.0128
NRMSE 0.024
NRMSE 0.194
NRMSE 0.182
NRMSE 0.112
NRMSE 0.145
NRMSE 0.006
NRMSE 0.168
NRMSE 1.08
NRMSE 0
NRMSE 0.524
NRMSE 1.84
NRMSE 0.969
NRMSE 0.969
NRMSE 14.9
NRMSE 5.93
NRMSE 0.963
NRMSE 0.87
NRMSE 0.4
NRMSE 0
NRMSE 0.00733
NRMSE 0.062
NRMSE 0.0801
NRMSE 0.14
NRMSE 0.743
NRMSE 0.04
NRMSE 0.0207
NRMSE 0.268
NRMSE 0.137
NRMSE 0
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
NRMSE 0.0217
NRMSE 0.0145
NRMSE 1
NRMSE 1
NRMSE 0.033
NRMSE 5.44
NRMSE 61.5
NRMSE 82
NRMSE 1.57e+03
NRMSE 0
NRMSE 0.311
NRMSE 0.158
NRMSE 1.04e+06
NRMSE 5.54e+06
NRMSE 0.195
NRMSE 1e+08
NRMSE 1.08e+04
NRMSE 14.9
NRMSE 52.4
NRMSE 0
NRMSE 4.52e-05
NRMSE 2.57e-05
NRMSE 0.937
NRMSE 0.922
NRMSE 5.37e-05
NRMSE 0.958
NRMSE 0.994
NRMSE 0.982
NRMSE 0.997
NRMSE 0
Cantera
NN mean of 3
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | REPAIRED
2 us self-fed steps; full 10 ms NRMSE = RMSE / RMS(Cantera); species axes symlog (1e-12)
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
Single-step baseline
Interval DeepONet
Self-fed multi-step
Cantera
NN mean of 3
NRMSE 0.000429
NRMSE 0.0243
NRMSE 0.0308
NRMSE 0.0346
NRMSE 0.00838
NRMSE 0.0137
NRMSE 0.00114
NRMSE 0.0425
NRMSE 0.0331
NRMSE 0
NRMSE 0.0294
NRMSE 0.141
NRMSE 0.208
NRMSE 0.664
NRMSE 0.234
NRMSE 1.02
NRMSE 0.0517
NRMSE 0.315
NRMSE 0.378
NRMSE 0
NRMSE 0.00787
NRMSE 0.192
NRMSE 0.11
NRMSE 0.0842
NRMSE 0.1
NRMSE 0.027
NRMSE 0.0176
NRMSE 0.17
NRMSE 0.108
NRMSE 0
Time (ms)